



A+ LAB SERIES V5

Lab 11: Sharing Resources - Printers

Document Version: **2026-01-02**

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	3.7: Given a scenario, deploy and configure multifunction devices/printers and settings.
CompTIA A+ (220-1202) Exam Objectives	1.7: Given a scenario, configure Microsoft Windows networking features on a client/desktop.

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Introduction

One of the fundamental purposes of a network is to share resources, such as printers. Understanding printer types and how to share them is a necessary skill for a PC technician.

Installing an individual printer on every computer would be extremely costly, in terms of funds, effort, and supplies. It is much more useful to centrally locate printers, share them on the network, and provide access to the computers that need them.

This lab will demonstrate how to install a local printer (one connected to either the parallel or USB ports on the computer) and a network-connected printer (one connected to the network using either a wired or wireless connection), configure printer sharing, and connect additional computers to the "printer share."

Objective

Installing an individual printer on every computer would be extremely costly, in terms of funds, effort, and supplies. It is much more useful to centrally locate printers, share them on the network, and provide access to the computers that need them.

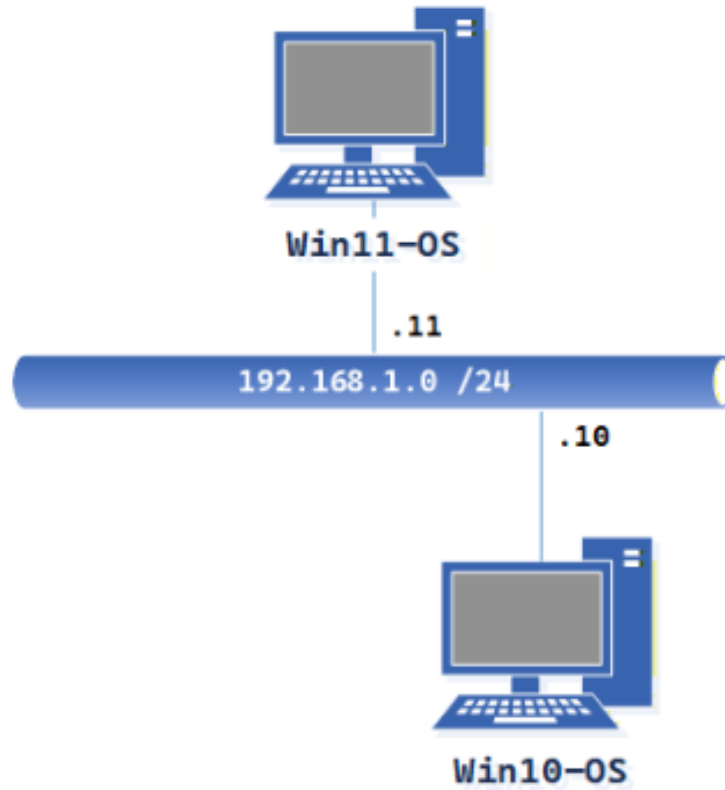
This lab will demonstrate how to install a local printer (one connected to either the parallel or USB ports on the computer) and a network-connected printer (one connected to the network using either a wired or wireless connection), configure printer sharing, and connect additional computers to the "printer share."

This lab provides hands-on experience installing and sharing printers in Windows 10 and Windows 11, and reviews key concepts of Windows printer sharing.

In this lab, we will be performing the following tasks:

- Installing and sharing a printer connected to the network
- Adding the new printer to a Windows client machine
- Install additional drivers

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Windows Computer Network Settings

This section of the lab will walk through the process of verifying *Windows-based* computer network settings.

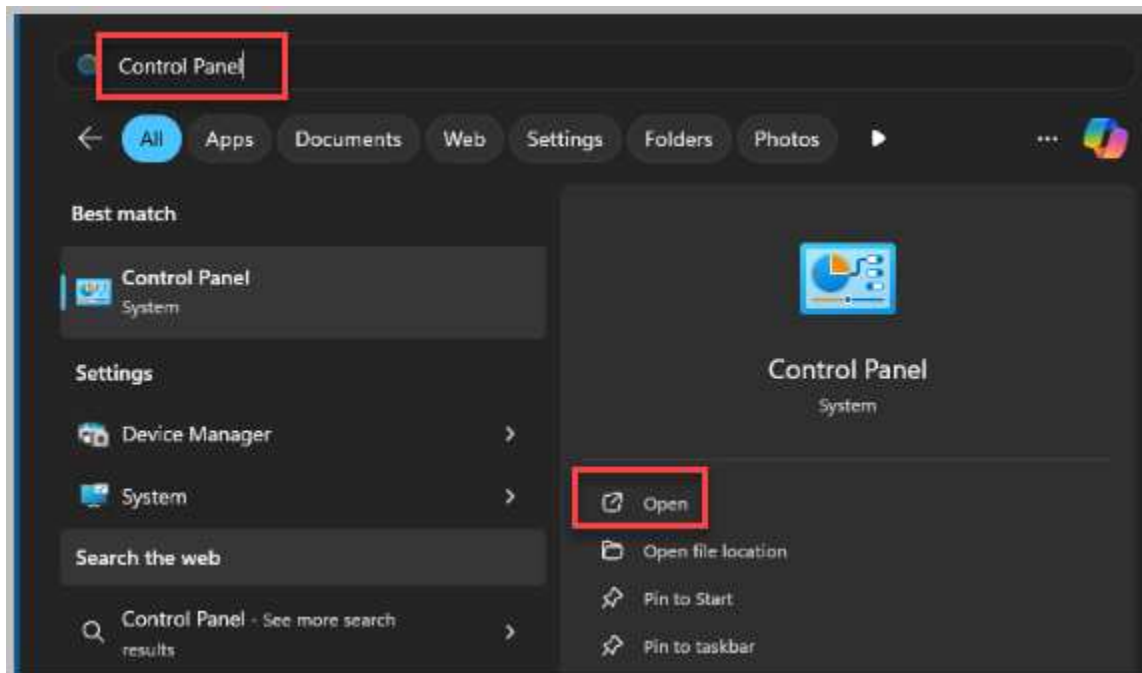
1.1 Win11-OS Computer Network Settings

1. Open a console connection to the **Win11-OS** system.

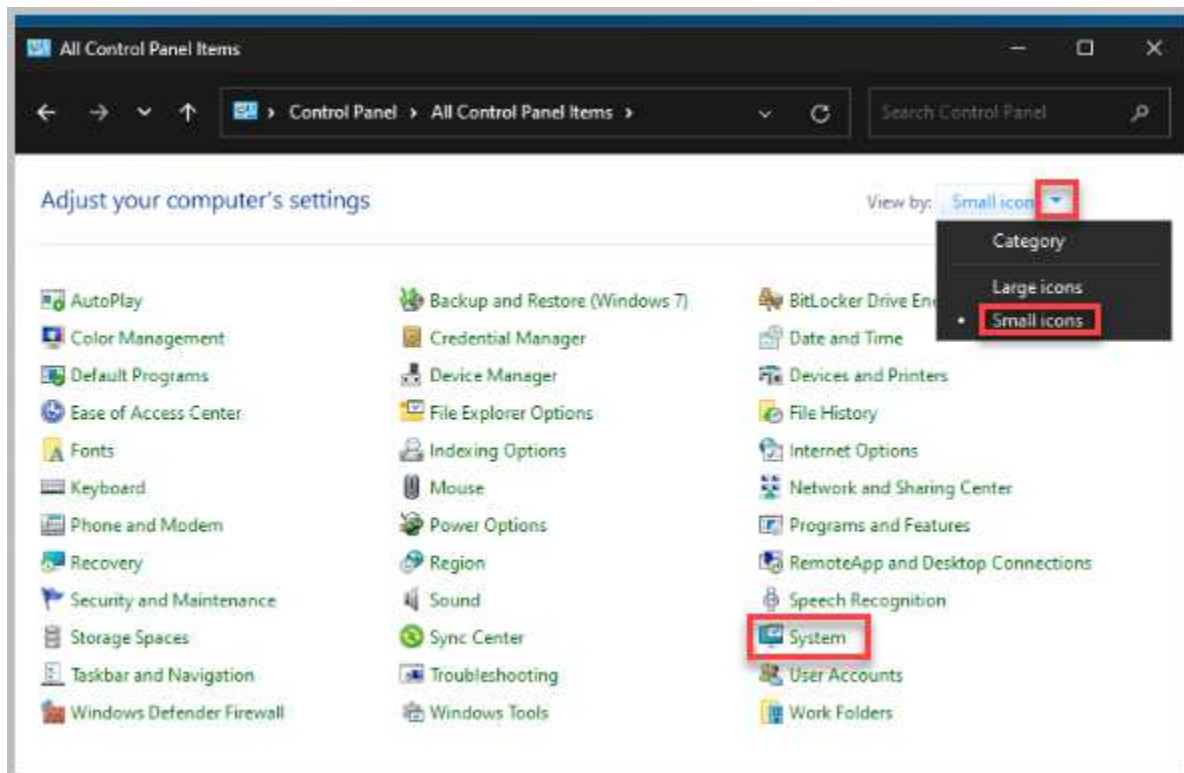


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

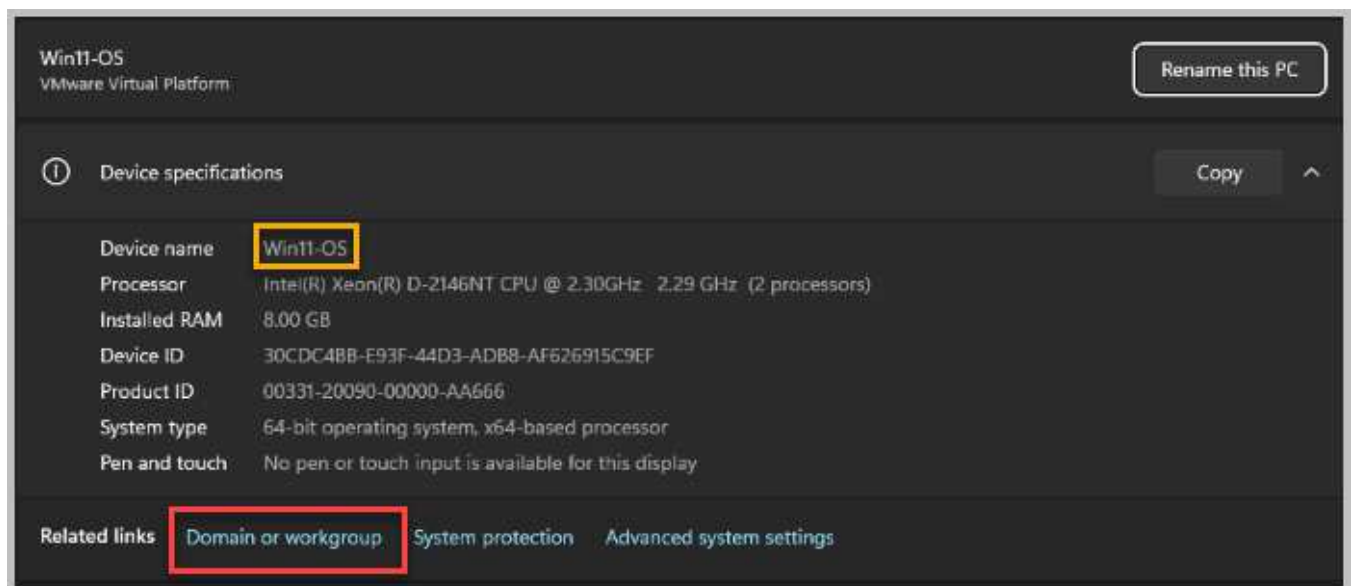
2. Select the search bar on the taskbar, type **Control Panel**, and select **Open**.



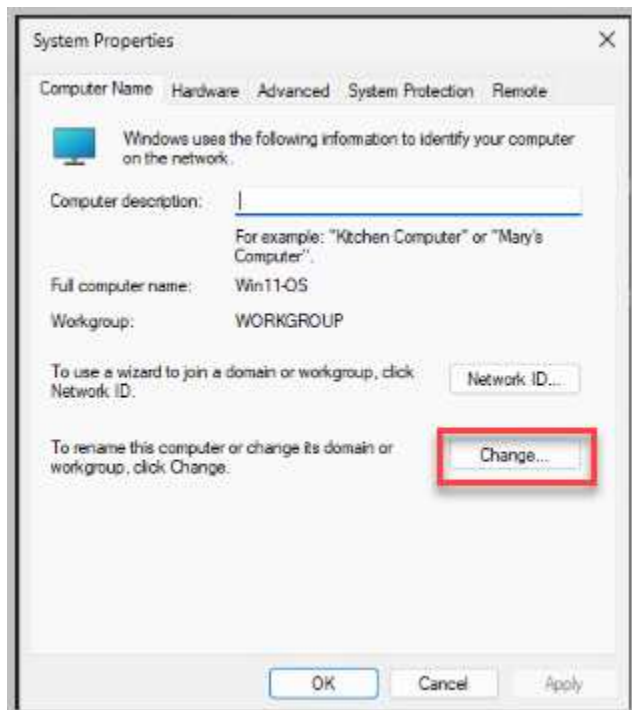
3. In the *Control Panel* window, change the *View* to **Small icons**. Click on **System**.



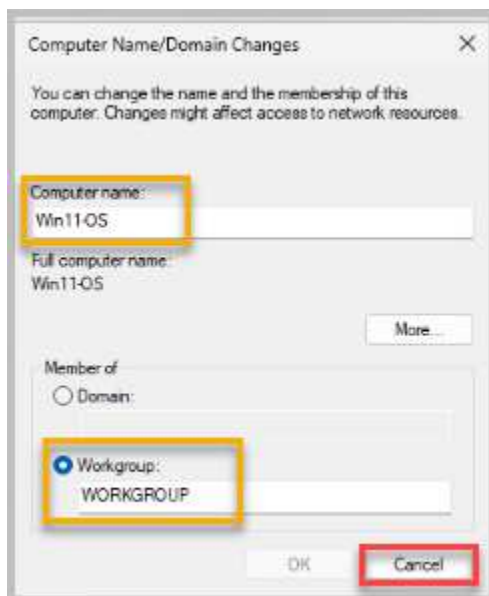
4. On the *System* window, locate the *Computer name* and select **Domain or workgroup**.



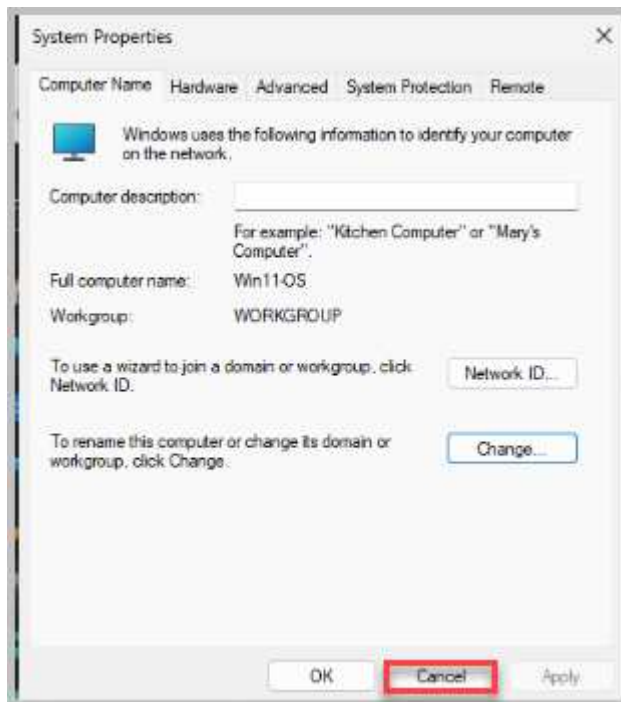
5. On the *System Properties* window, click the **Change** button.



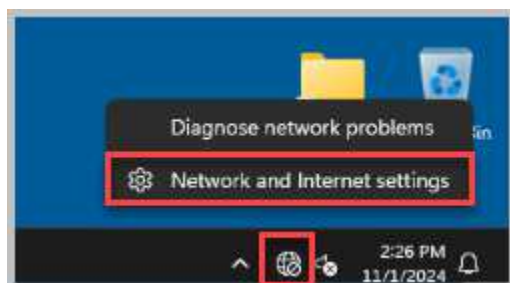
6. Verify the Computer name is **Win11-OS** and the *Win11-OS* System is part of the **WORKGROUP** workgroup. Click **Cancel**.



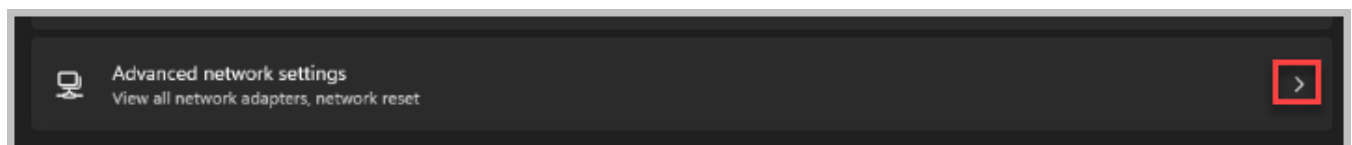
- Click **Cancel** to close the *System Properties Window*.



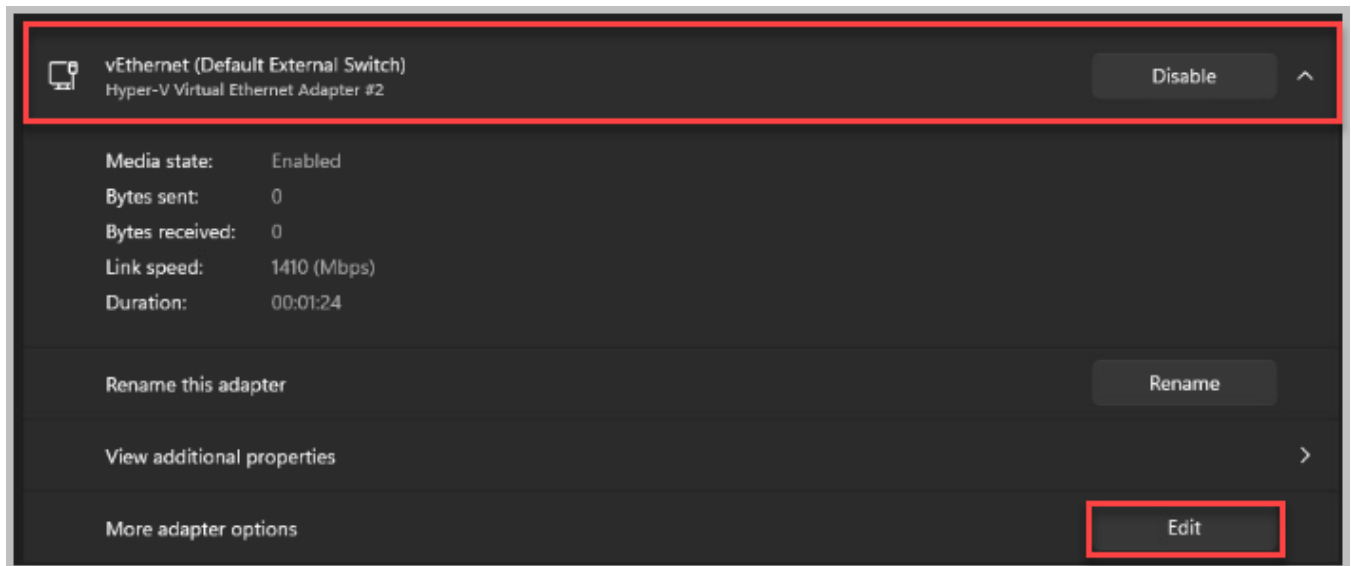
- Close the *About System* and *Control Panel* windows.
- Right-click on the **Network** icon (located in the notification area) and then click on the **Network and Internet settings** menu option.



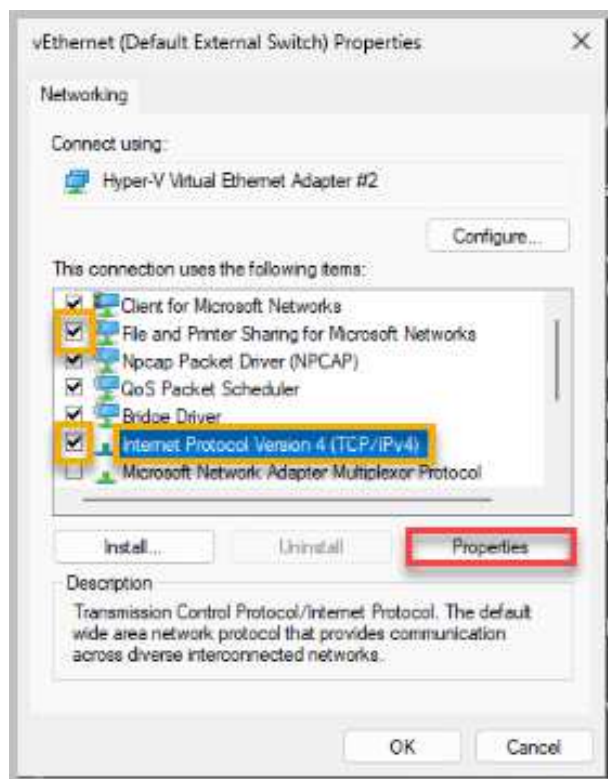
- Select **Advanced network settings**.



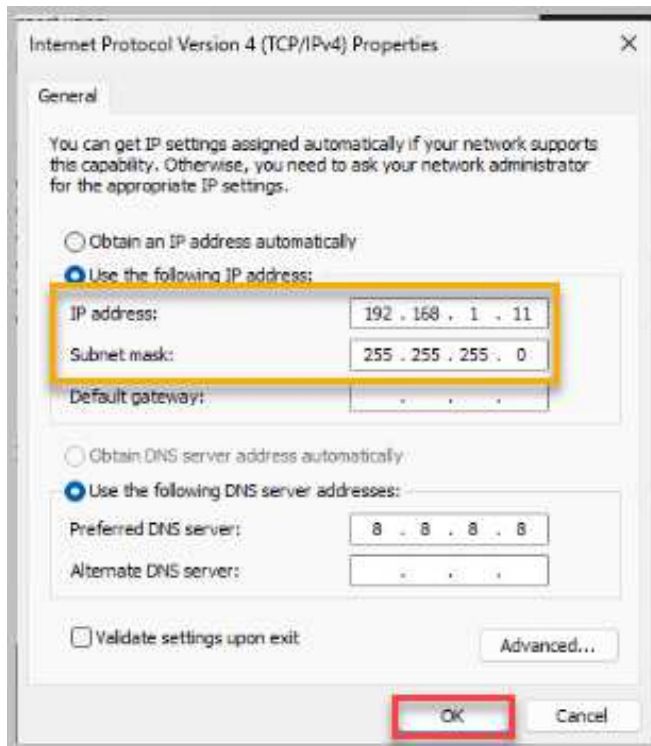
11. Select the **vEthernet (Default External Switch)** adaptor. Then select **Edit** next to *More adapter options*.



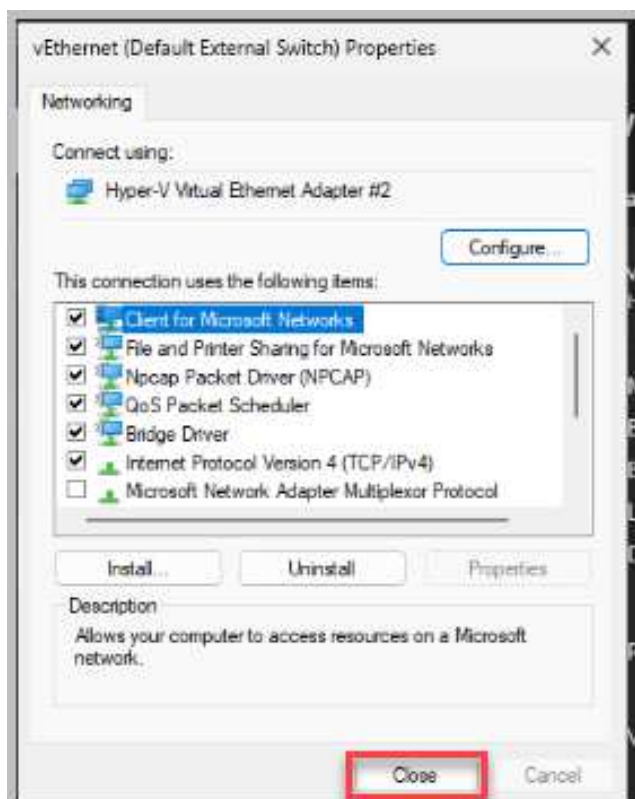
12. Ensure the **File and Printer Sharing for Microsoft Networks** and **Internet Protocol Version 4 (TCP/IPv4)** checkbox is selected. Make sure **Internet Protocol Version 4 (TCP/IPv4)** is highlighted. Select **Properties**.



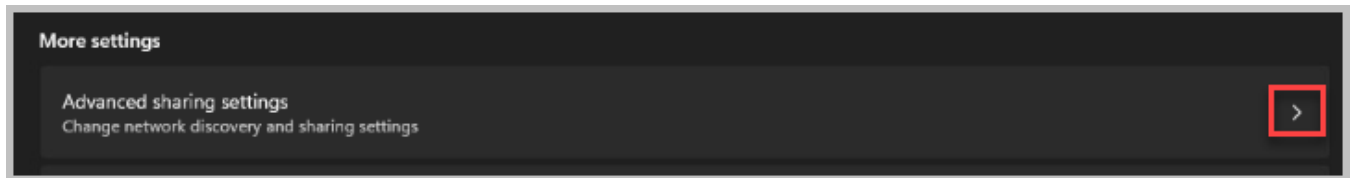
13. Verify the Static IP Address is set, as you will need this address later in this lab. Select **OK** to close the IPv4 Properties Window.



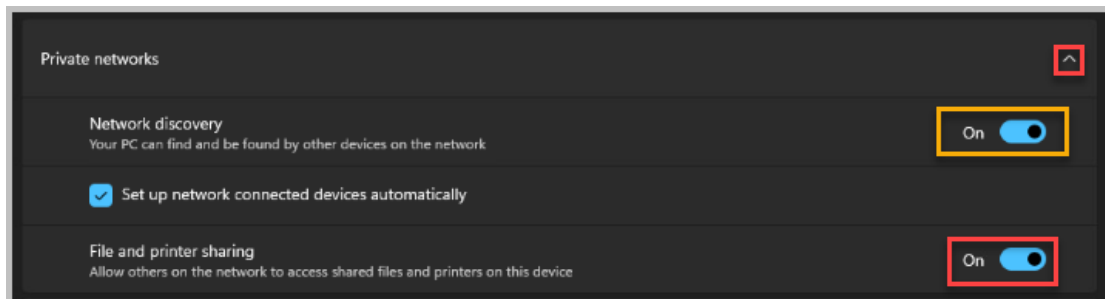
14. Select **Close** to close the *vEthernet Properties* window.



15. Click on **Advanced sharing settings**.



16. This window allows you to change the sharing options for various network profiles. Expand *Private networks* by selecting the **drop-down arrow**. Ensure that **Network discovery** is **On** and turn on **File and printer sharing** radio button.



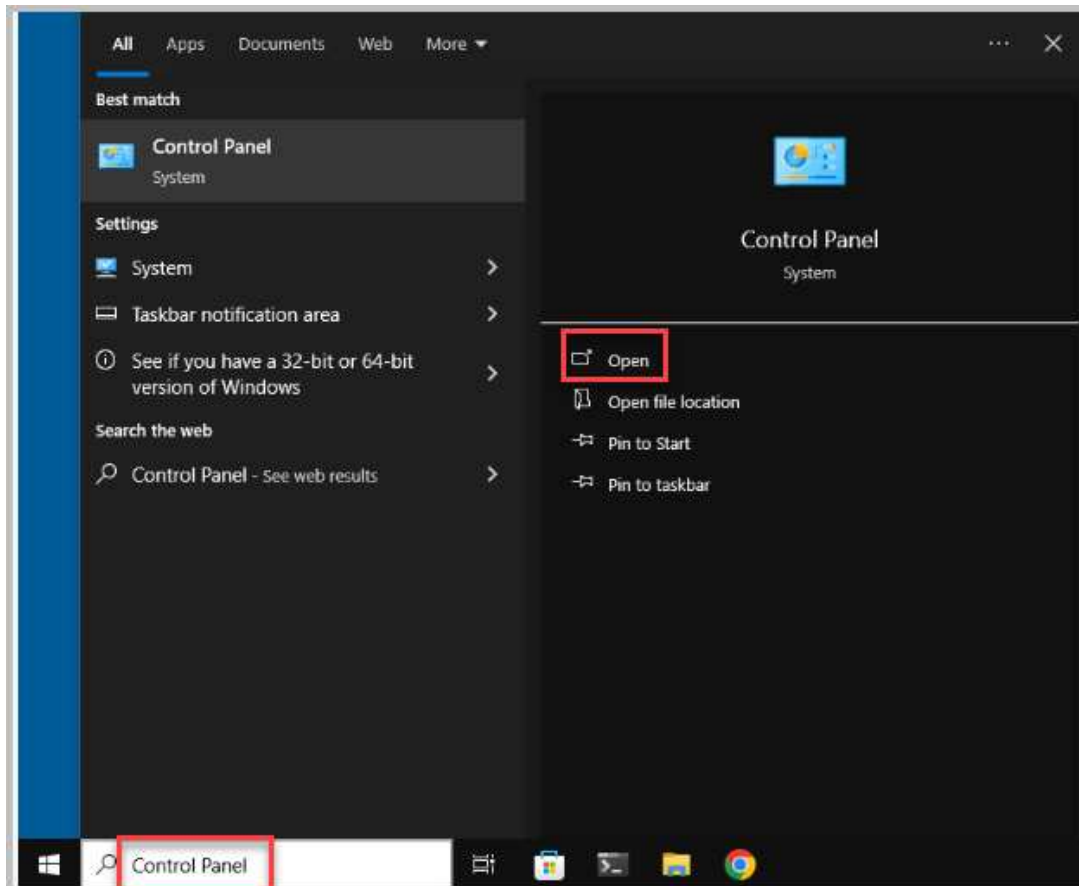
17. Close the *Advanced Network Settings* window.

1.2 Win10-OS Computer Network Settings

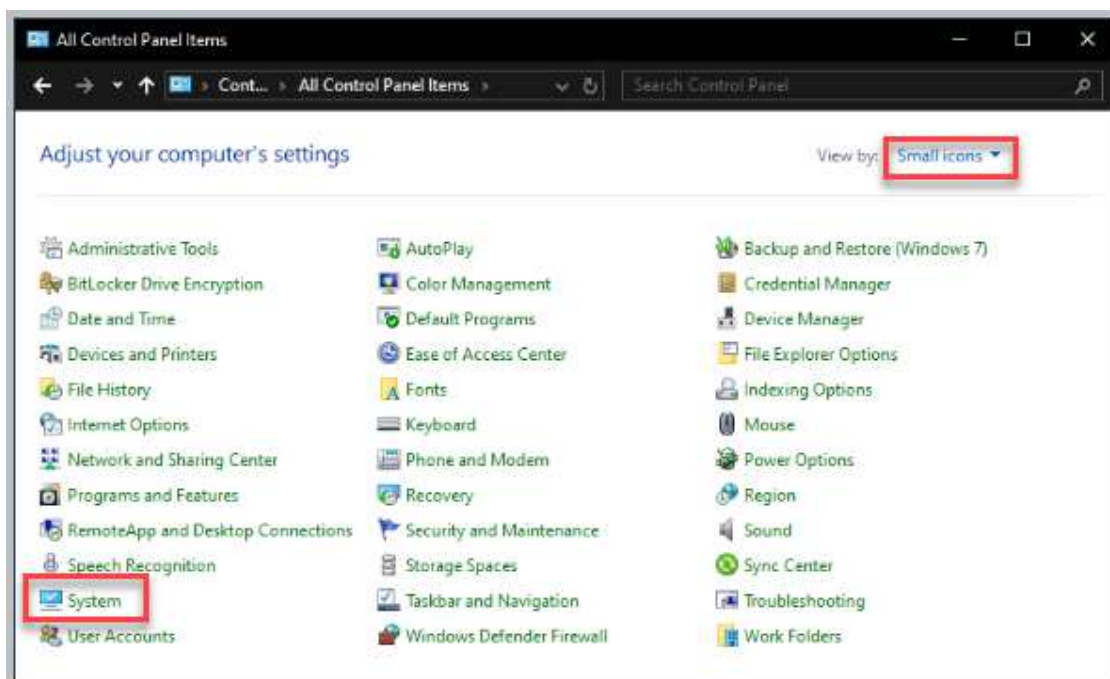
1. Switch to the **Win10-OS** system.



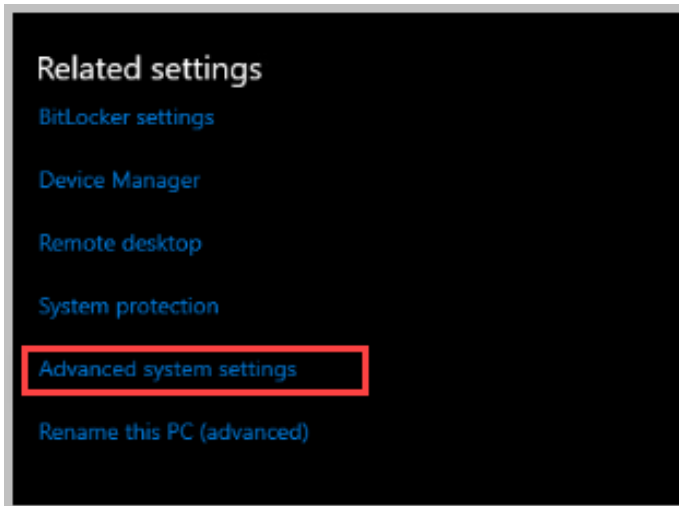
2. Select the search bar on the taskbar, type the command **Control Panel**, and select **Open**.



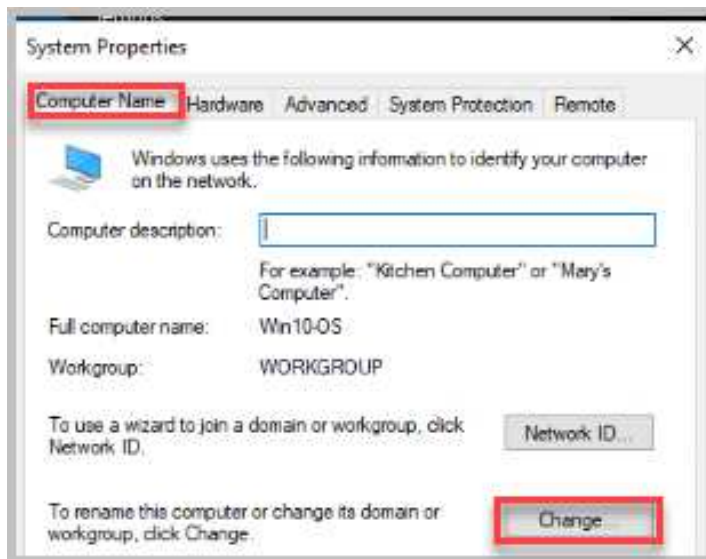
3. In the *Control Panel* window, ensure that **Small icons** are set for the *View by* and click on **System**.



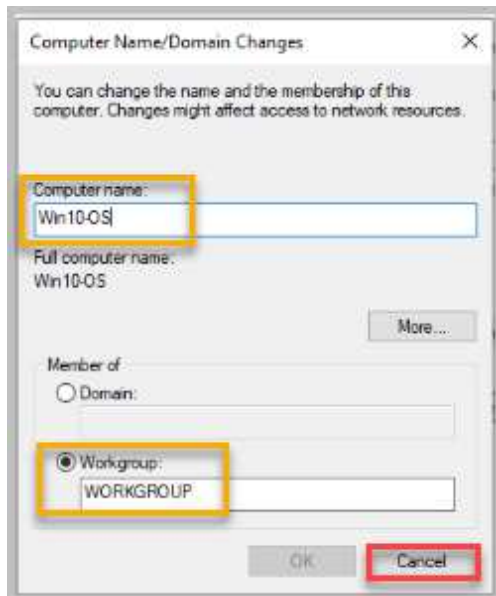
4. In the right side of the *About* window, scroll down to *Related settings* and click on the **Advanced system settings** link.



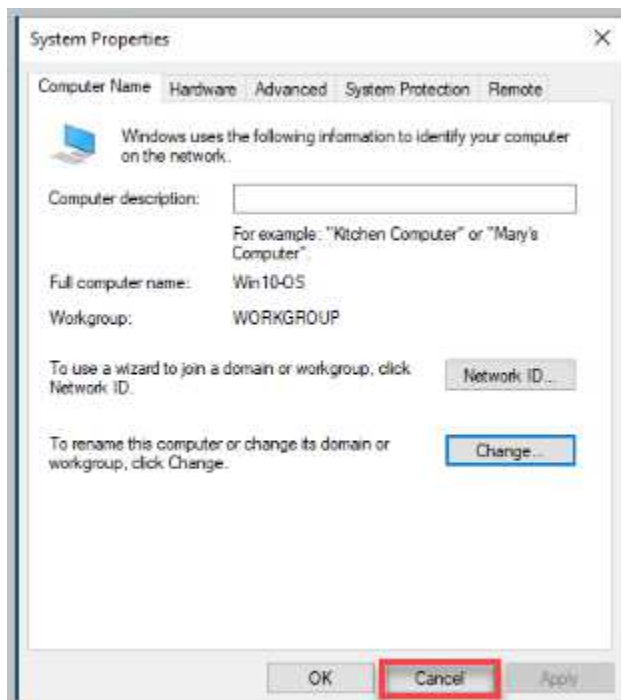
5. Select the **Computer Name** tab and select **Change**.



6. Verify the Computer Name is **Win10-OS**, and the *Win10-OS* System is part of the **WORKGROUP** workgroup. Click **Cancel**.

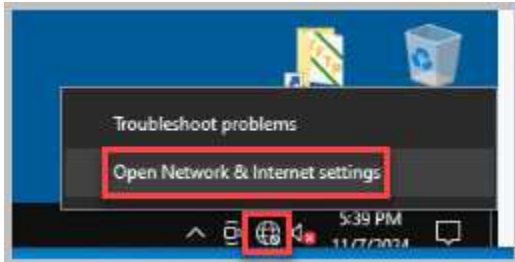


7. In the *System Properties* window, click the **Cancel** button.

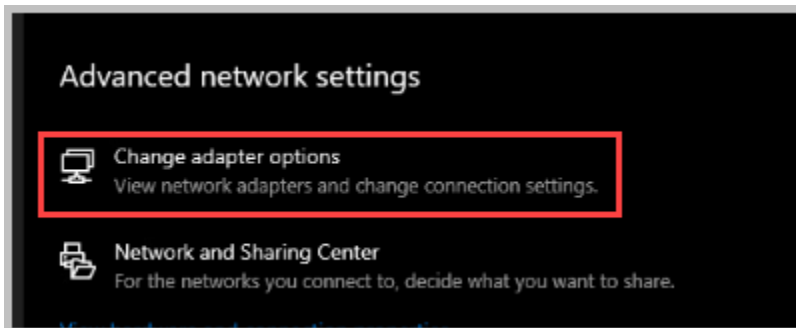


8. Close the *About System Settings and Control Panel* windows.

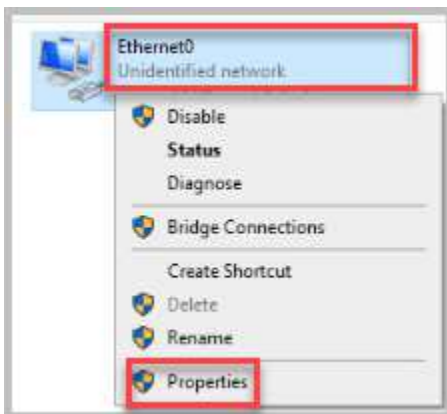
9. Right-click on the **Network** icon (located in the notification area) and then click on the **Open Network & Internet settings** menu option.



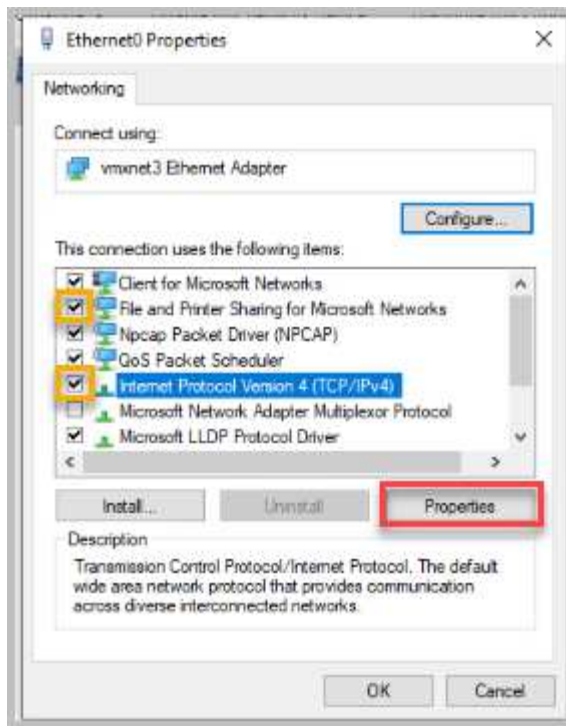
10. Under *Advanced network settings*, click **Change adapter options**.



11. Right-click on **Ethernet0** and click on **Properties**.



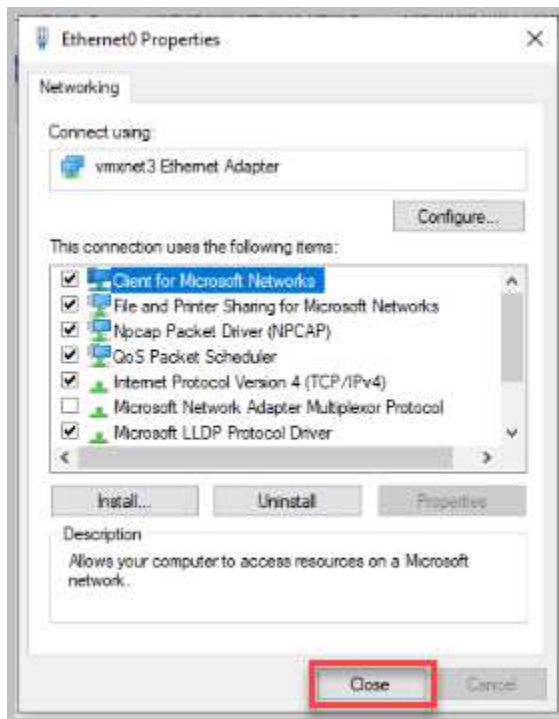
12. Ensure the **File and Printer Sharing for Microsoft Networks** and **Internet Protocol Version 4 (TCP/IPv4)** checkbox is selected. Make sure **Internet Protocol Version 4 (TCP/IPv4)** is highlighted. Select **Properties**.



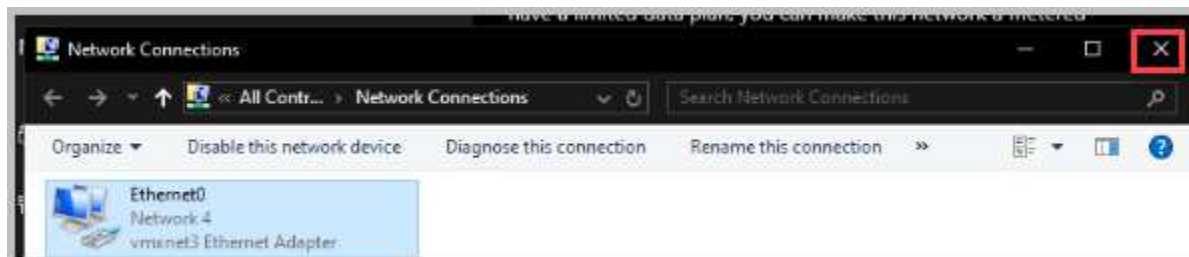
13. Verify the Static IPv4 Address Settings. Select **OK** to close IPv4 Properties window.



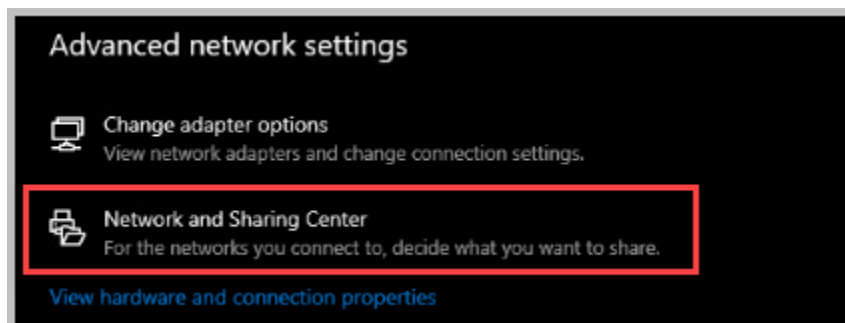
14. On the *Ethernet0 Properties* window, select **Close**.



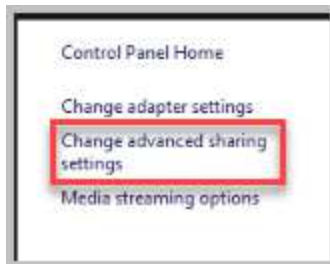
15. Close the *Network Connections* window.



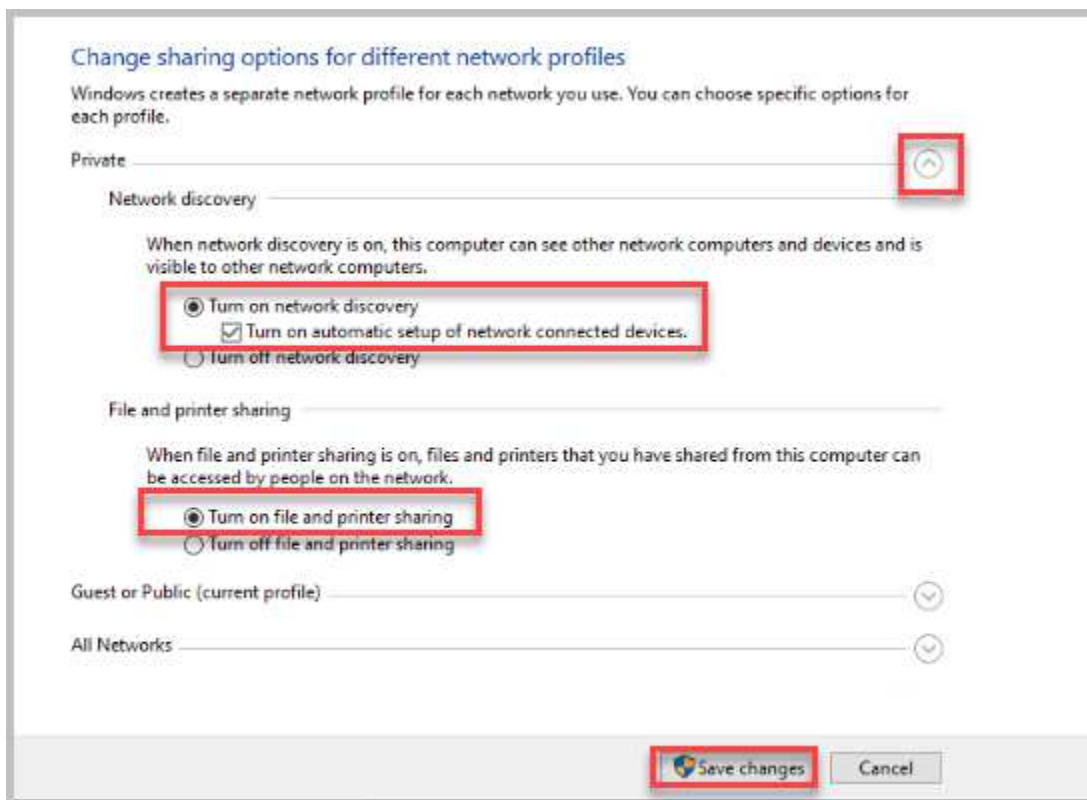
16. Under *Advanced network settings*, click on **Network and Sharing Center**.



17. Select **Change advanced sharing settings**.



18. Select the **Drop-Down Arrow** for the Private Network. Ensure that **Turn on network discovery** and **Turn on file and printer sharing** radio buttons are selected. Click **Save changes**.



19. Close all open Windows.

2 Local Printer Setup

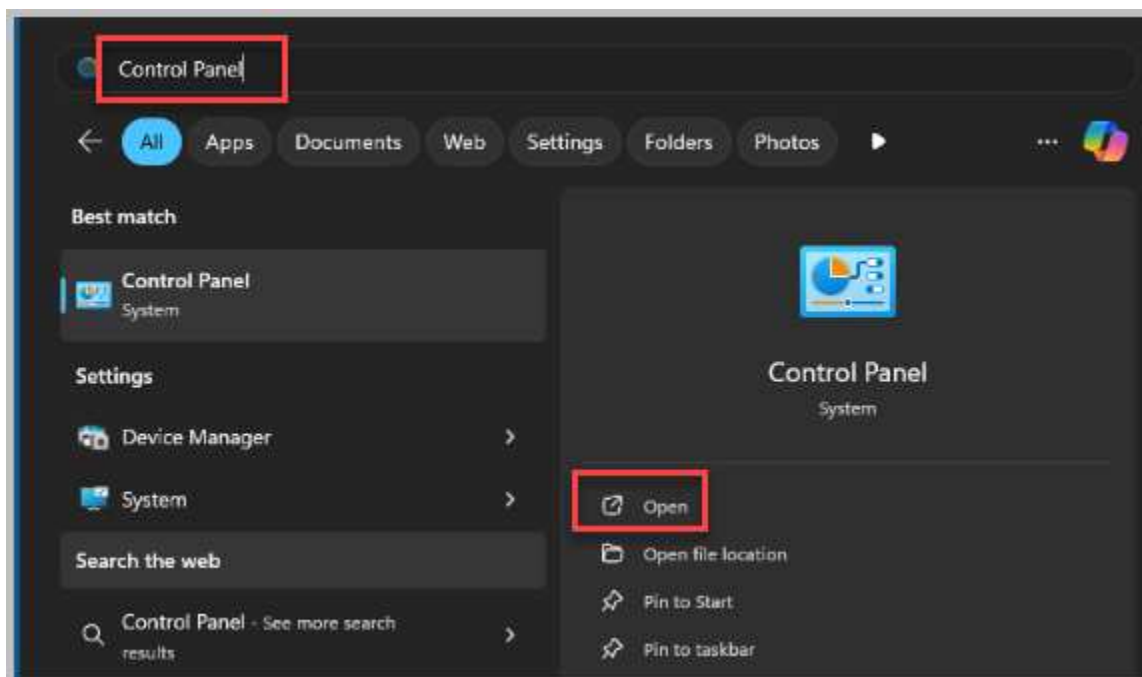
Both versions of *Windows* have different methods to access the printer setup and configuration utility. This task will outline the steps required for the Win10 OS.

2.1 Adding a Local Printer in Win10-OS

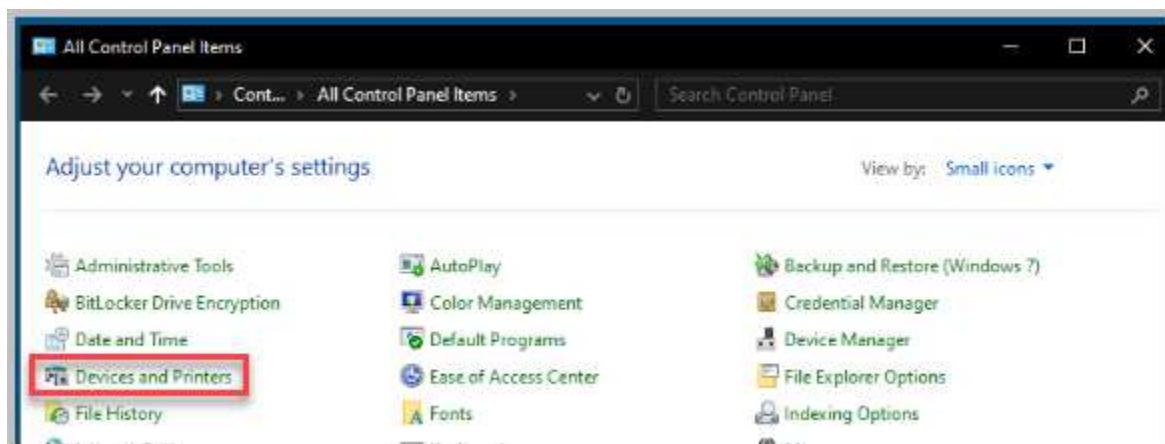
1. Open a console connection to the **Win10-OS** system.



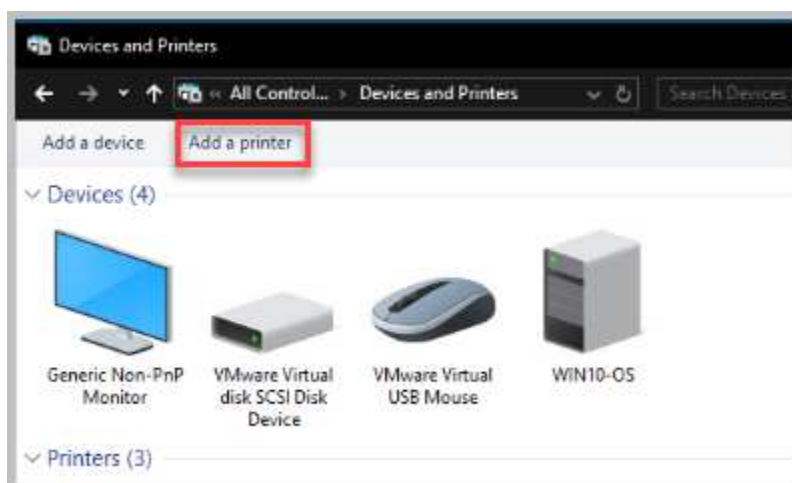
2. Select the search bar on the taskbar, type the command **Control Panel**, and select **Open**.



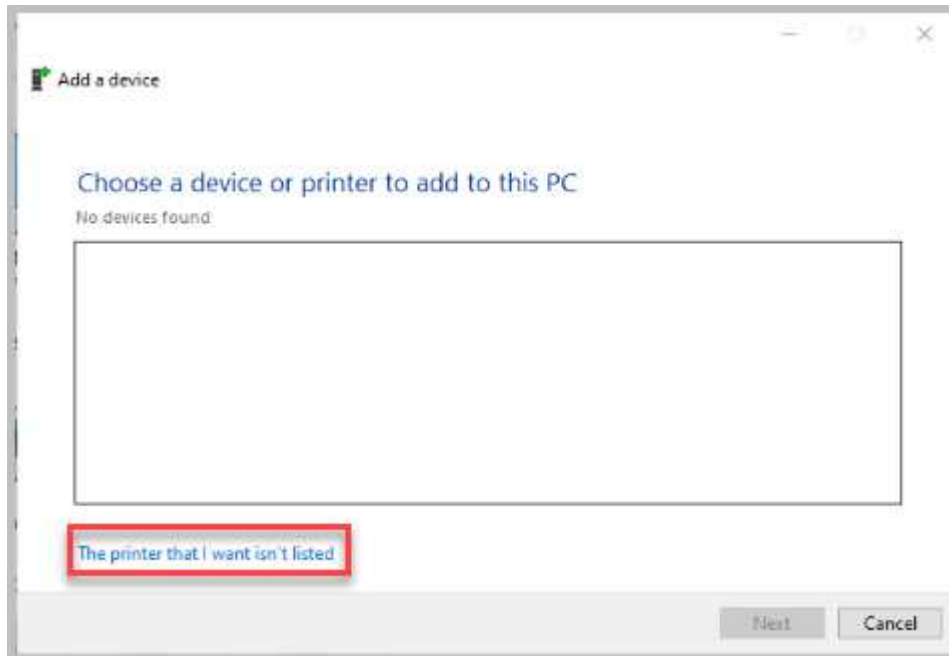
3. In the *Control Panel* window, click on **Devices and Printers**.



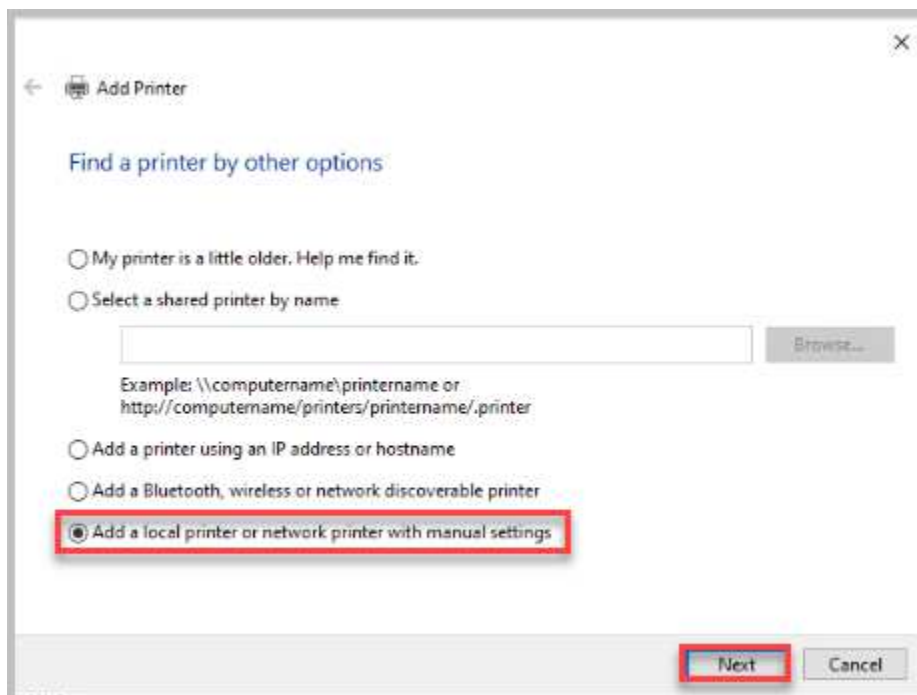
4. Notice the currently available devices. Click on **Add a printer**.



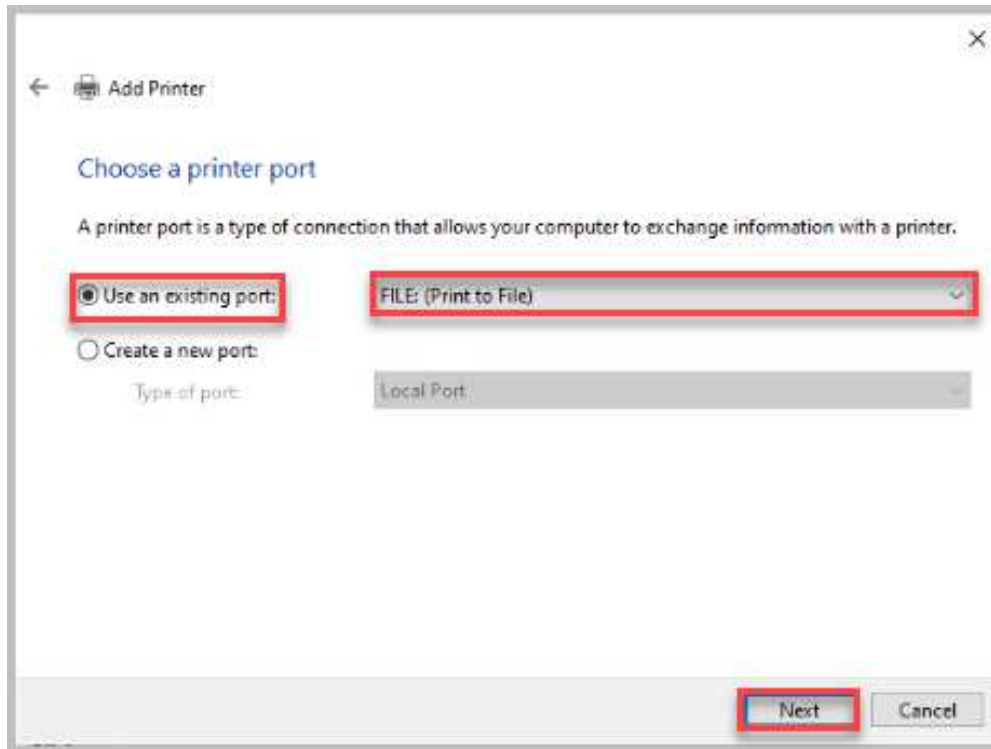
5. The *Add Printer* window will open and will automatically search for shared printers that are already installed on the network. Once the application finishes searching (no printer will be found), click on **The printer that I want isn't listed.**



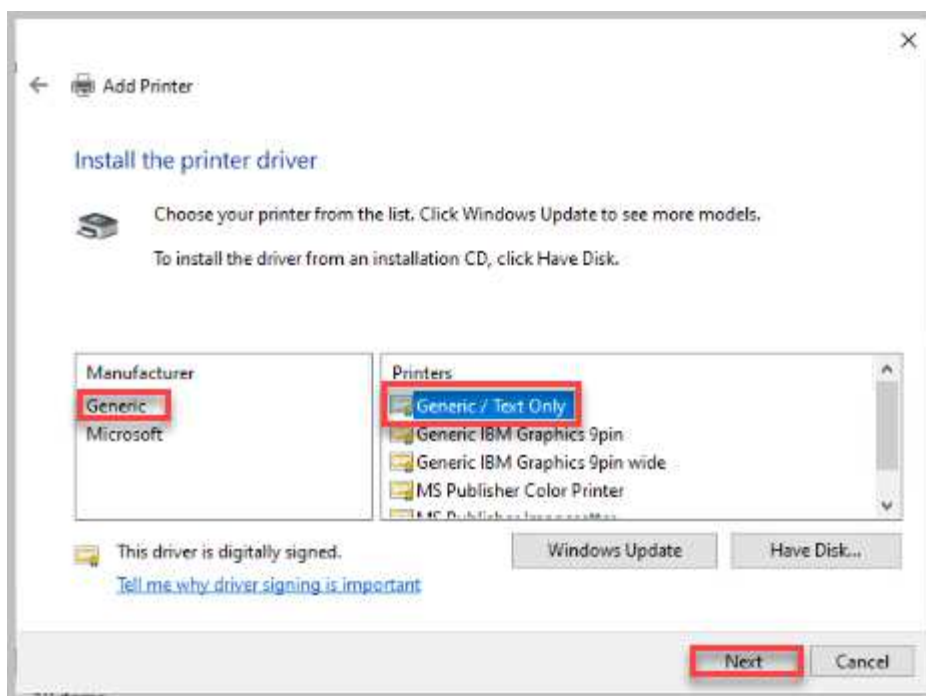
6. Select the radio button for **Add a local printer or network printer with manual settings.** Click **Next.**



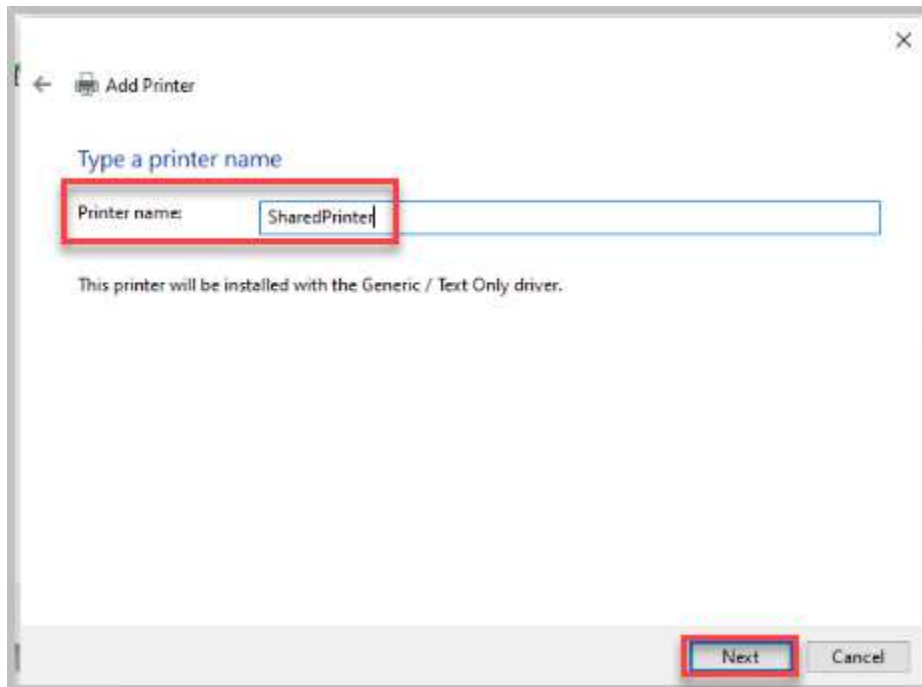
7. In a real-world situation, you may encounter legacy equipment, and if the printer you intend to install connects using a legacy parallel port (*LPT*) or even a *USB* port, then “*Use an existing port*” would be selected. If the printer has a network interface card (*NIC*), then “*Create a new port*” would be chosen. Due to the virtual environment, you will select **Use an existing port** and select **FILE: (Print to File)** from the dropdown menu. Click **Next** to continue.



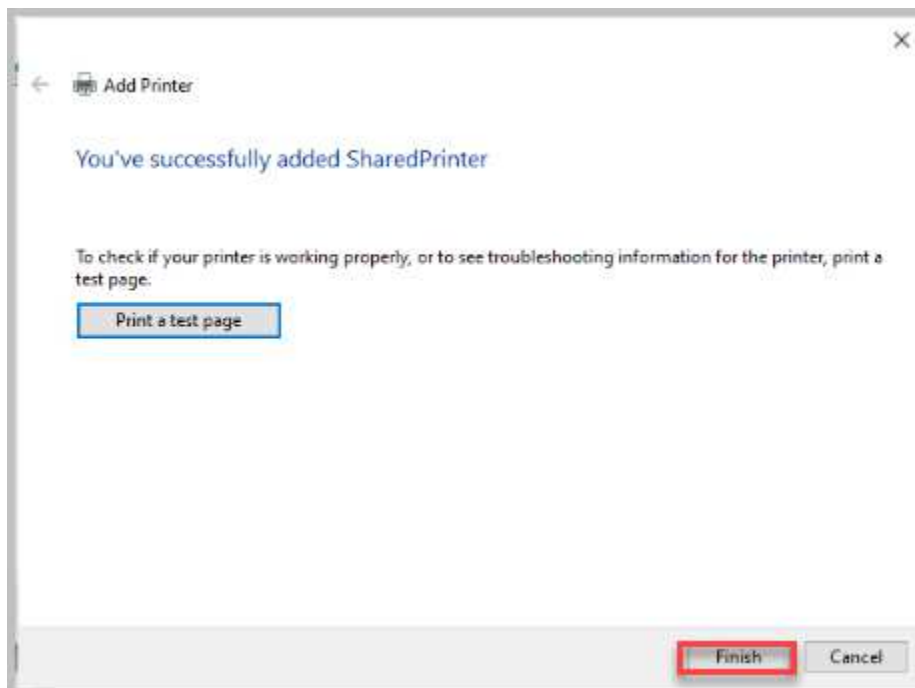
8. The next step of the wizard will ask you to choose a printer driver. Select **Generic** underneath the *Manufacturer* pane, followed by selecting **Generic / Text Only** from the *Printers* pane. Click **Next**.



9. For the Printer Name, type **SharedPrinter**. Select **Next**.



10. Select **Finish** to complete the Add Printer Setup.



Normally, you would want to print a test page, so you would click on the *Print a Test Page* button. Since there is no physical printer, we will not be printing a test page.

11. Leave the *Devices and Printers* window open.

3 Network Printer Setup

This task will highlight the steps required for sharing a locally configured printer to be accessible on a network in the same workgroup.

3.1 Sharing Network Printer from Win10-OS System

A printer can be connected directly to the network using wired or wireless connections. A printer connected in this manner is not considered a “shared” printer. It is regarded as a “local” printer, but the interface is a NIC with an IP address. Therefore, we only have to add the network location as the “port”.

1. Verify you are consoled into the *Win10-OS* System.



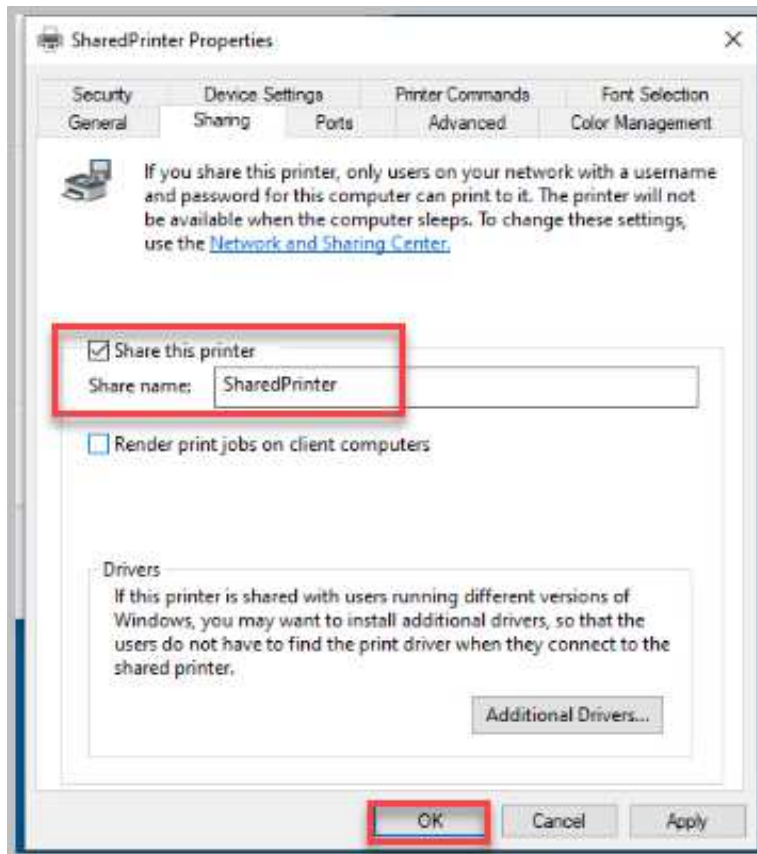
2. In the *Devices and Printers* open window, click the new **SharedPrinter** and select **Printer properties**.



3. Select the **Sharing Tab** and select **Change Sharing Options**.



4. Select **Share this printer** and leave the default name of **SharedPrinter**. Select **OK**.



3.2 Connecting Network Printer to the Win11-OS System

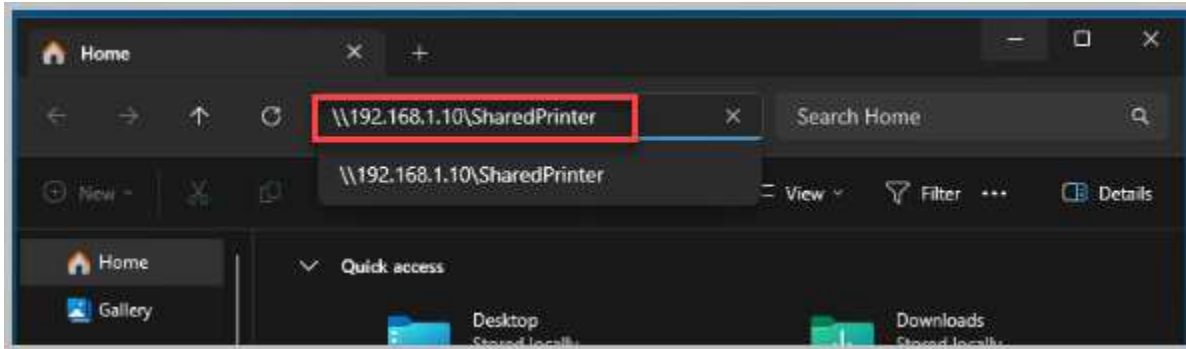
1. Open a console connection to the **Win11-OS** system.



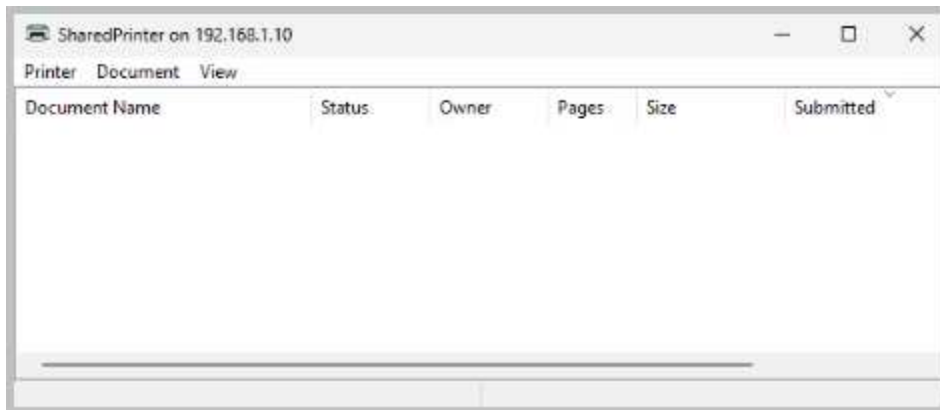
2. Select **File Explorer** in the taskbar.



3. Type `\\192.168.1.10\SharedPrinter` in the network address of file explorer and select **Enter** on your keyboard.



4. The *SharedPrinter* will auto-connect to your Win11-OS System.



5. The lab is now complete; you may end the reservation.